LocalLift

Turning local news into a café's next good day.

An AI research agent that reads the neighbourhood's news, events and buzz, then tells a small café what's coming, why it matters and what to do about it.

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Software Engineering Management · Fall 2026

LIVE PROTOTYPE · VERCEL
locallift-pied-mu.vercel.app

SOURCE CODE · GITHUB
github.com/sumeethaldipur/LocalLift



PROJECT REPORT · NEWSREN BRIEF

OCTOBER 27, 2026

1 THE PROBLEM

A small café that keeps finding out too late

Meet Maya. She runs Opening Act Café, two blocks from Madison Square Garden in Midtown Manhattan.

Some evenings, a wave of concert-goers pours through the door an hour before a show. The line goes out the door, the pastry case is empty by 6pm, and Maya is the only one on the till. She loses sales simply because she can't serve people fast enough.

Other days she stocks up for a crowd that never comes: the game was moved, a street was closed for a parade, or the subway line was down.

> **"Everything that affects my day is in the news somewhere. I just never see it in time, and I don't have time to look."**

Why this is a burning problem

1

The signals are scattered

Event listings, local news, city permits, subway alerts and social posts all live in different places. Nobody has time to check ten sites every morning.

2

News doesn't say "so what"

A headline says "Harry Styles plays MSG." It doesn't say "expect about 80 extra customers between 5 and 7pm."

3

Timing is everything

Knowing about a crowd the morning of is too late. You need days to schedule staff, order stock and plan a special.

4

Plans change quietly

Cancellations, time changes and transit disruptions are reported in passing, often after you've already prepared.

The cost of not knowing

Small food businesses run on thin margins. One missed rush, or one over-prepared quiet night, can wipe out a week's profit. Big chains have analysts for this. Independent cafés don't.

What's real here: the café is made up so we could show the idea end to end. Everything around it is real: events from Ticketmaster and NYC Parks, news from Gothamist, Google News and GDELT, MTA transit alerts, NYC street permits, and social buzz from Bluesky and X.

2 THE IDEA

Local news as an early-warning system

The Newsren brief asks a simple question: what if news didn't just inform you, but told you what to do? We applied that to one person who really needs it: a café owner.

News is the earliest public signal that something will change on your street. A concert announcement, a road closure permit, a "show postponed" story, a subway service change. LocalLift reads all of it, connects it to the café, and turns it into a decision.

From headline to action

1 · Read
News, events, permits, transit alerts and social posts near the café

›

2 · Connect
Group stories about the same thing into one "development"

›

3 · Verify
Check against the official listing; flag what disagrees

›

4 · Explain
Why it matters to this café: crowd, timing, distance

›

5 · Act
A specific step: staff up, run a special, or change nothing

How news shows up inside the product

N

Local news & alerts

Street events, transit changes and neighbourhood stories, each with a suggested move.

S

Story intelligence

Many articles about one thing become one evolving story with a "what changed?" history.

E

News per event

Each event shows only the coverage that's actually about it, and checks it for cancellations or time changes.

B

Social buzz

How much people are talking about an event on X and Bluesky, and whether it's rising.

Q

Follow a question

"Tell me if this game is called off." The agent keeps checking the news for you.

A

Ask the news

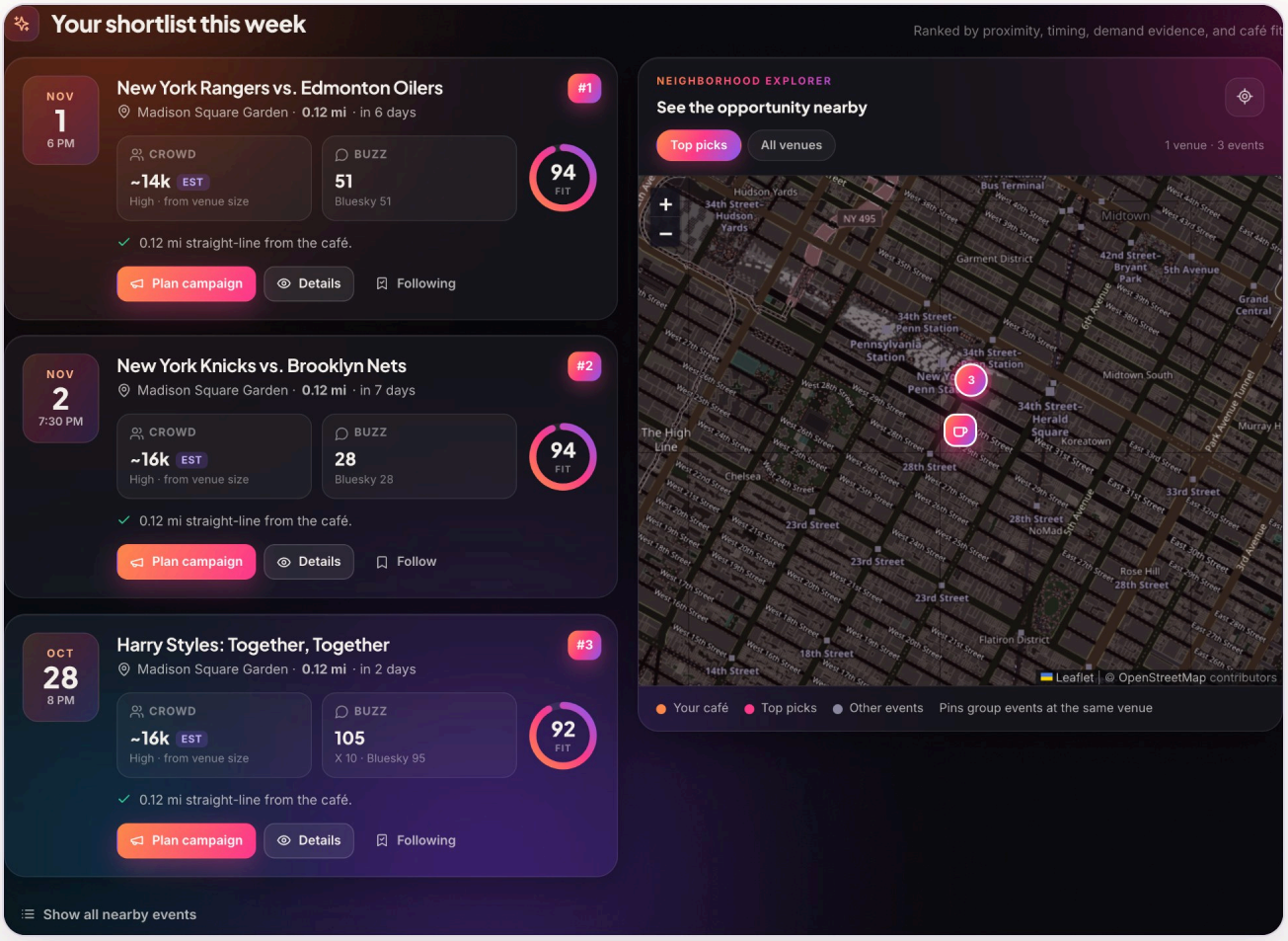
Ask anything about an event and get an answer built from cited sources.

Keeping the news relevant. Early on, the Harry Styles page showed articles about sneakers and Marvel films, because those stories happened to mention him. We added a rule: an article only counts if the event is in the headline, or the headline plus the text clearly talks about the show or venue. Now the news tab only shows coverage about the show.

3 MEET LOCALLIFT

One screen that answers “what’s coming?”

Over 1,700 events happen within two miles every fortnight. Maya doesn’t need 1,700. She needs the three that matter, and she gets them with her morning coffee.



EVENTS **Top picks this week, on a live map.** Each event gets a plain, explainable score for distance, timing, crowd size, demand and fit with the café’s hours. The café is pinned in orange, the best opportunities in pink.

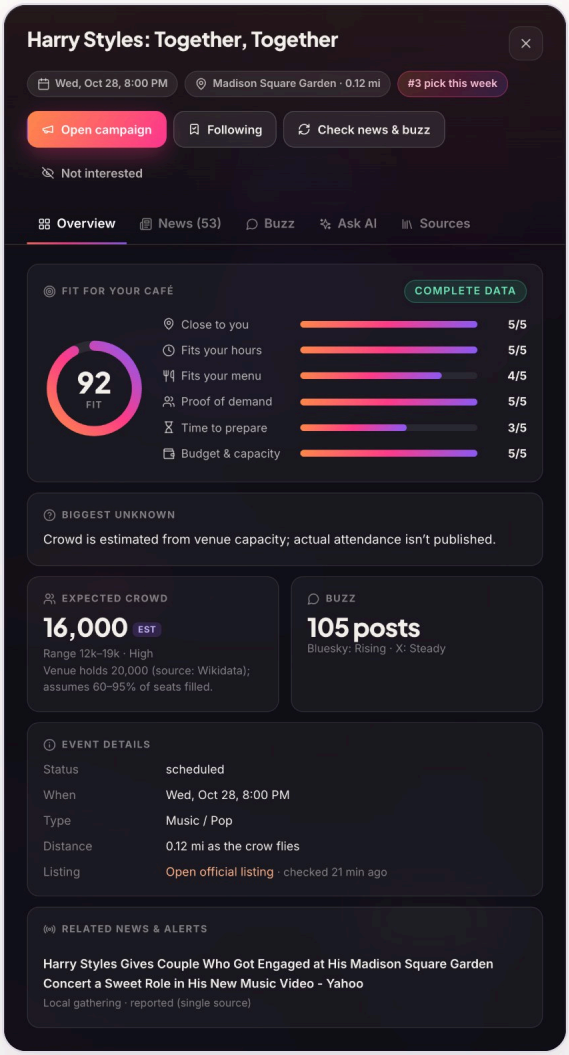
A typical morning, in five taps

1. **Read the briefing:** the week’s top picks and anything urgent, like a cancellation or a subway closure.
2. **Open an event:** the expected crowd, the distance, the relevant news and the buzz.
3. **Ask the agent:** “Is this still on, and when will people pass the café?” It answers with sources.
4. **Plan a campaign:** a themed special and a step-by-step plan, with the profit worked out.
5. **Check Rush Radar:** which hours will be busy, and whether she needs an extra pair of hands.

Every screenshot in this report carries a tag for the capability it shows: NEWS AGENT EVENTS ACTION TRUST

Inside an event: why it matters

Tapping an event opens its full picture: the crowd, the timing, what the news says and exactly what to do.



EVENTS **Event overview** for Harry Styles at Madison Square Garden, 0.12 miles away.

How to read this panel

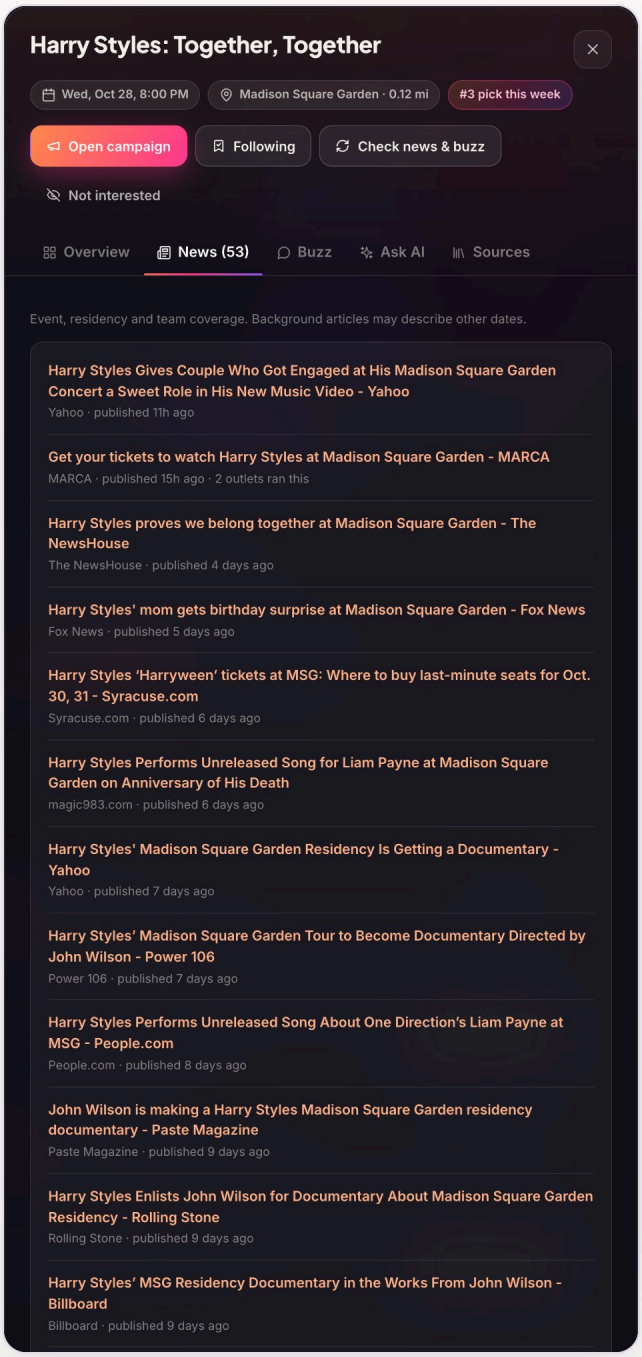
- **Fit score (92)**
  How well this event suits this café: close by, during opening hours, matches the menu, strong demand, enough time to prepare, within budget.
- **Biggest unknown**
  The one thing we couldn't confirm. Here: real attendance isn't published, so the crowd is estimated.
- **Expected crowd: about 16,000**
  The arena holds 20,000 (from Wikidata), and a typical show fills 60 to 95% of seats. Always marked "EST".
- **Buzz: 105 posts**
  Live chatter on Bluesky and X, with a trend for each.
- **Related news**
  Stories linked to this event, labelled with how well they're confirmed.
- **Official listing**
  A link to the source, and when we last checked it.

Crowd estimates, never "unknown"

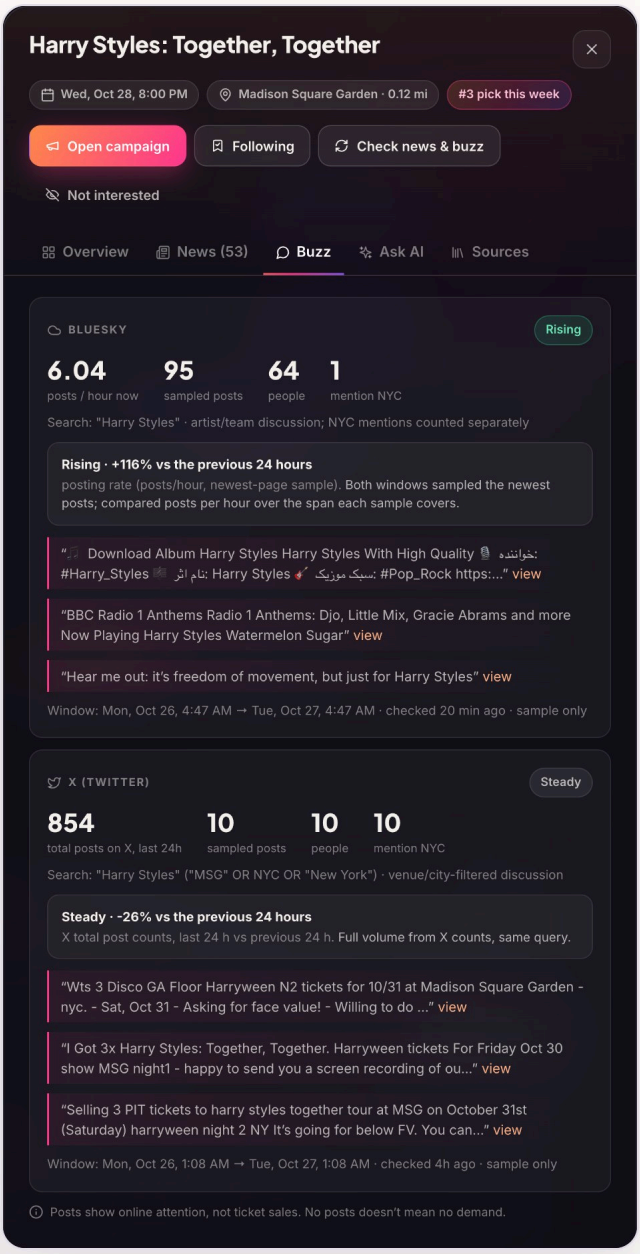
A published figure if there is one; otherwise venue capacity times typical sell-through; otherwise a ballpark for that type of event. Always labelled as an estimate.

4 THE FEATURES

The news and the buzz around each event



NEWS **Event news.** Only coverage that is really about this show, each item linked to its source and dated.



NEWS **Social buzz.** Post counts on X and Bluesky shown separately, with a trend that compares today with yesterday.

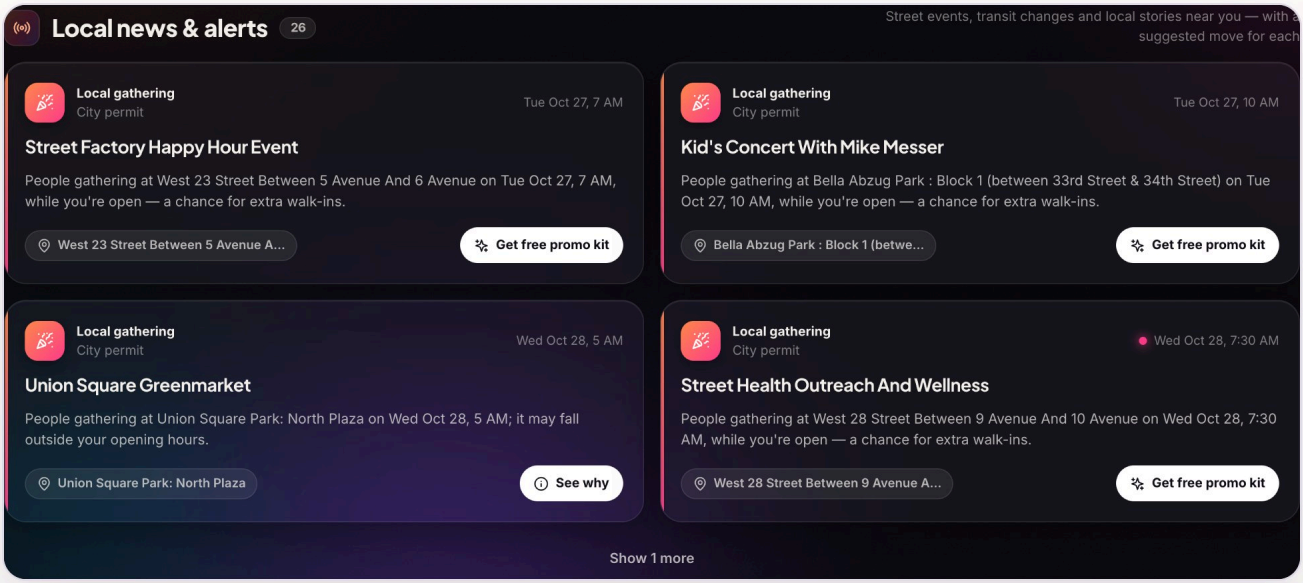
Why buzz matters

A sold-out show and a half-empty one look the same on a ticket listing. Rising chatter is an early hint that more fans will show up, or that something has changed. X is checked once a day for the top two events, under a $5 a month cap that Maya approved. Bluesky is free and checked more often.

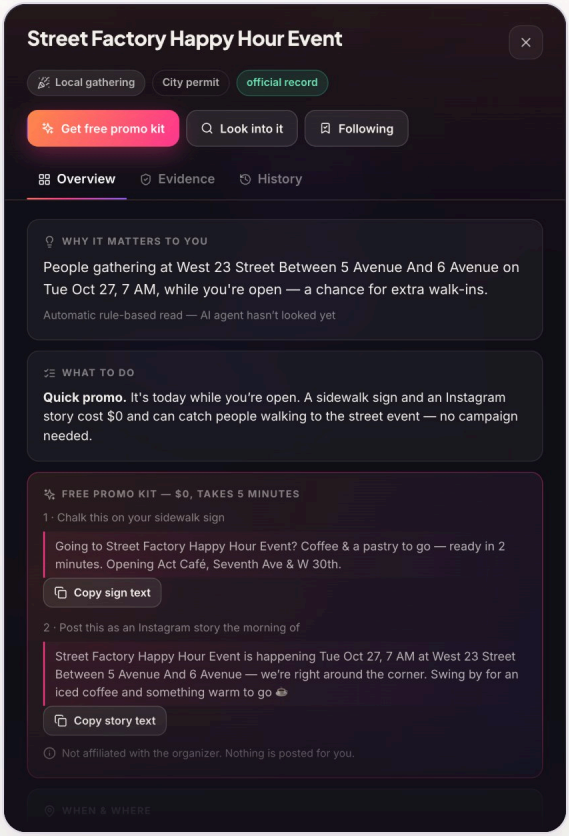
4 THE FEATURES

Local news that tells you what to do

Not every important change is a concert. Street fairs, parades, closures and subway work matter too.



NEWS **Local news & alerts.** Stories from Gothamist, Google News, NYC permits and MTA alerts, grouped into developments. Each card says what happened, why it matters to the café and a suggested move.



NEWS **One development, opened.** A city permit for a street event nearby, turned into a $0 promo.

From a permit to a plan

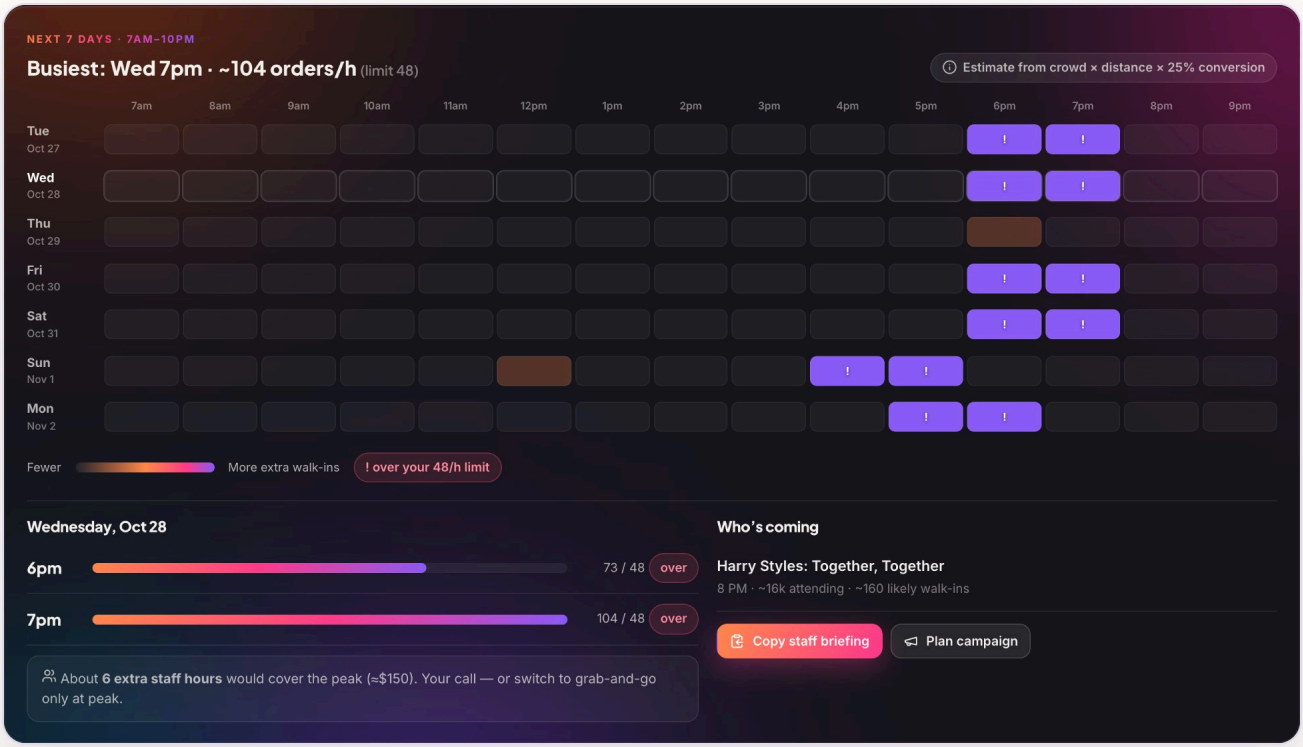
This one started as a dry city record: a curb-lane closure for a street event on West 23rd Street. LocalLift noticed it's close by, happens while the café is open, and could bring people past the door.

- **Why it matters to you**
  Written for this café, not for a general reader.
- **What to do**
  A specific move sized to the opportunity. Here, a quick promo, because it's today and small.
- **Free promo kit**
  Ready-to-copy text for a sidewalk sign and an Instagram story. Costs $0, takes 5 minutes. Nothing is posted for you.
- **Evidence and history**
  Every source behind the story, and a version history of what changed since Maya last looked.

4 THE FEATURES

Rush Radar: when to expect the wave

A seven-day, hour-by-hour view of extra walk-ins from every nearby event, stacked on top of normal business.



EVENTS **Rush Radar.** Each column is an hour of the day (4pm, 5pm...). The brighter the cell, the more extra customers. A "!" means the café would be over its limit of orders per hour.

How it's worked out

Crowd size × the share of people who pass the café (based on distance) × the share who come in. The walk-ins are spread over the two hours before the event starts, because that's when people grab a coffee on the way.

Tap a day and LocalLift writes a short **staff briefing**: which events are on, when it gets busy, what to prep, and whether an extra staff hour would pay off.

Staff briefing · Wednesday Oct 28

Harry Styles at MSG, 8:00 PM (~16,000 attending, ~160 likely walk-ins)

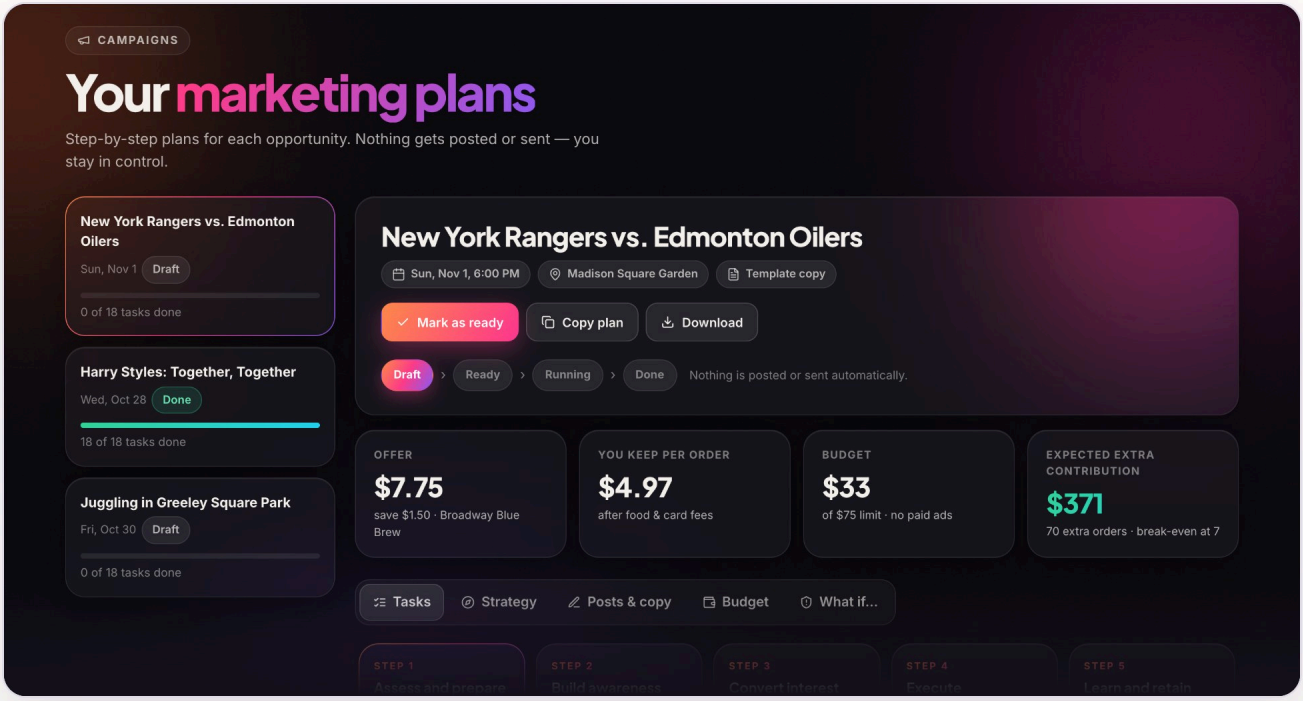
6pm: ~73 orders expected (normal ~8), over our 48/h limit
7pm: ~104 orders expected, over our limit by ~56

Prep: pre-wrap grab-and-go; cold brew pre-poured for the peak.
Owner decision: ~6 extra staff hours (≈$150), or grab-and-go only at peak.
Script: "Heading to the show? We're ready in two minutes."

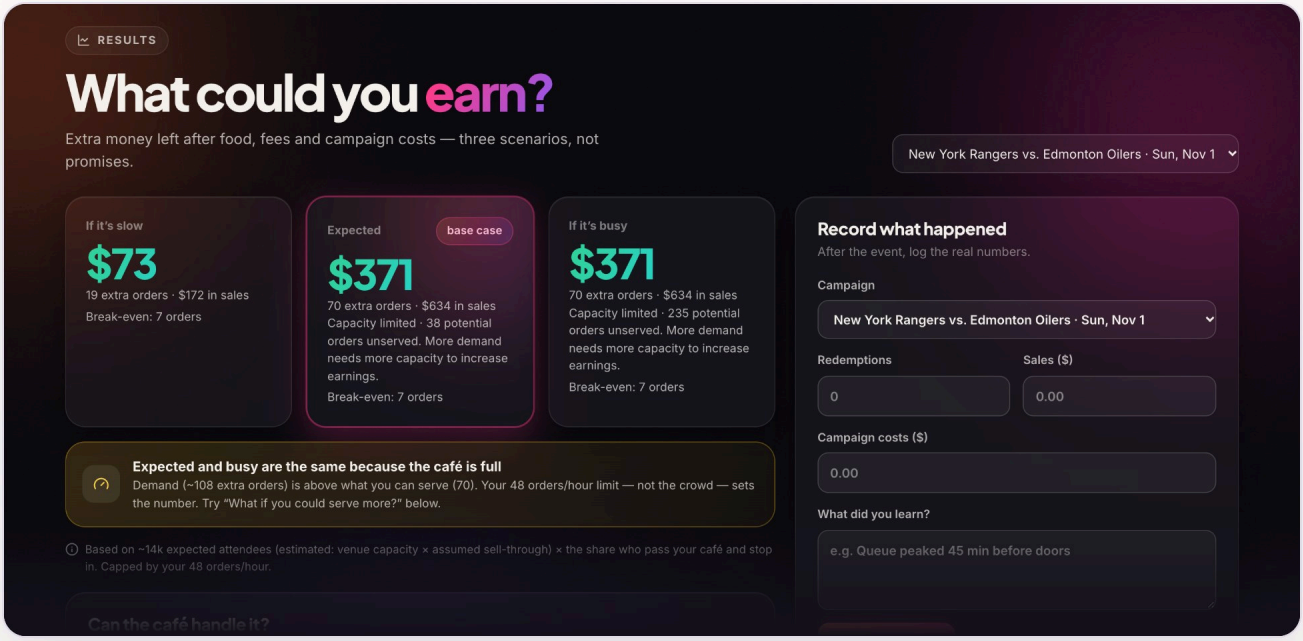
ACTION The real staff briefing LocalLift wrote for Wednesday, one tap to copy.

4 THE FEATURES

From insight to a plan that pays



ACTION **Campaign plan.** A themed special made from the café's real menu (here, the "Broadway Blue Brew" at $7.75; no artist names or lyrics), ready-made sign and social copy, and five dated steps. Nothing is posted or sent automatically.



ACTION **What could you earn?** Slow, expected and busy scenarios after food, fees and campaign costs, worked out by code, not the AI. When demand is more than the café can serve per hour, it says so, so Maya doesn't overspend on promotion she can't serve. After the event she records what really happened.

5 HOW IT IS AGENTIC

Not a chatbot. A researcher with a job.

The brief is clear: calling an AI to write a summary is not agentic. An agent needs a goal, the right tools and a controlled way of gathering evidence. Here is how LocalLift does each.

🎯 A clear goal

Every run has one job: investigate an event, answer Maya's question, judge a news development, scan the week, or keep watch on a question.

🧰 Real tools

It decides which of 14 tools to use, and in what order:

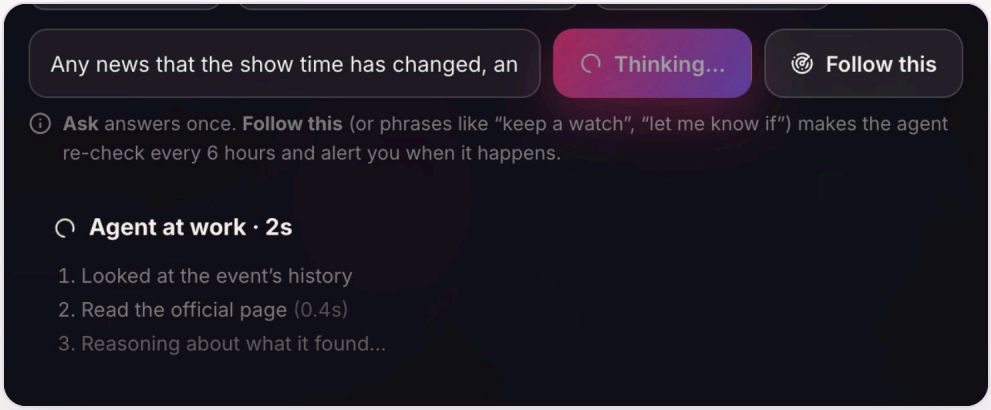
`discoverEvents` `searchNews` `searchSocial` `fetchApprovedSource` `getEventHistory` `getDevelopmentHistory` `calculateDistance` `rankOpportunities` `calculateDemand…` `recordClaim` `savePlanDraft` `saveBriefingDraft` `createWatch` `getBusinessProfile`

🛡️ A controlled process

At most 8 look-ups, 60 seconds and 12 steps per run. Every tool call is checked. Only approved websites. Numbers come from code, not from the AI's imagination.

What a run looks like

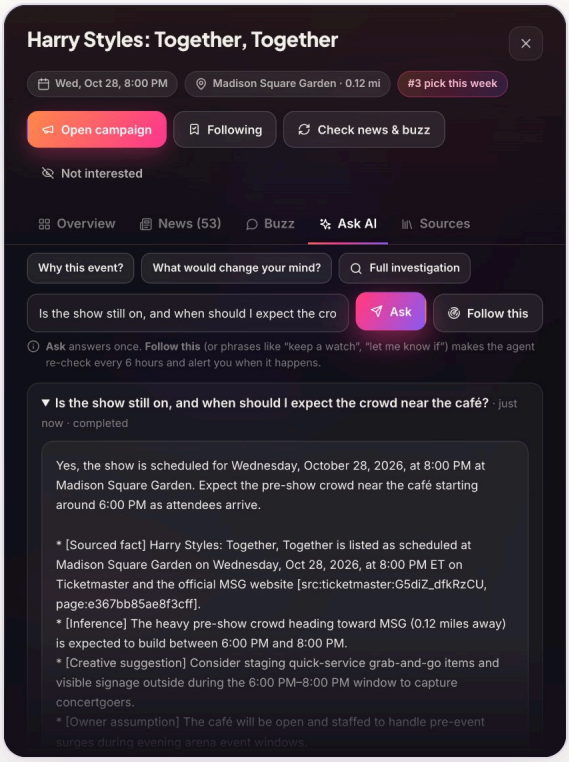
Plan "To answer this I need the listing and the latest news." › **Act** Calls tools: listing, news, buzz, official page › **Observe** Reads the results; notices gaps or conflicts › **Adapt** A source refuses access? Tries the official one instead › **Report** Labelled answer with sources and what's still unknown



AGENT **The agent at work, live.** While it runs, Maya sees each step it has finished and how long it took: it is choosing tools, not just writing text.

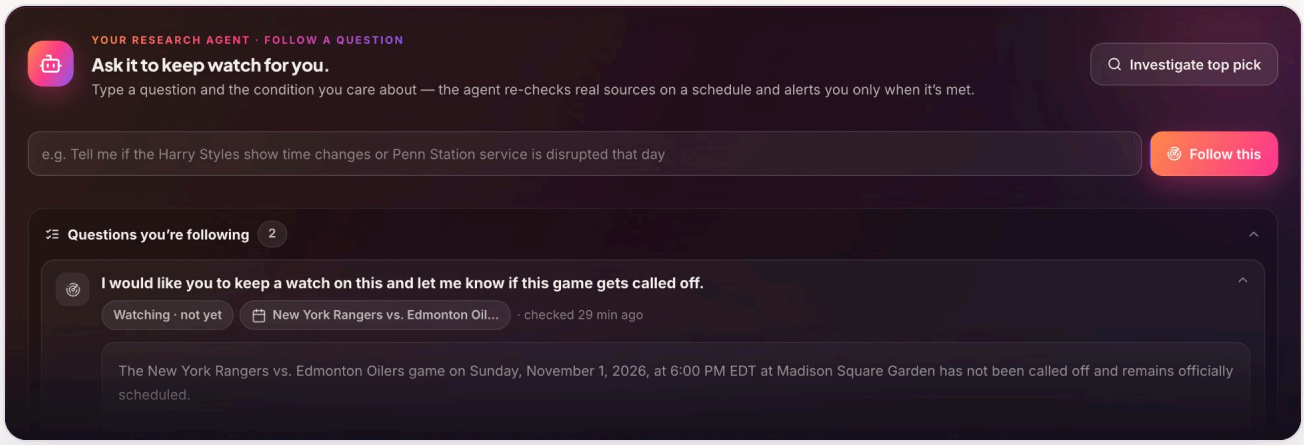
5 HOW IT IS AGENTIC

Answers you can check, and a question it keeps watching



AGENT **A grounded answer**, about 5 seconds after asking.

- **A direct answer first**
  "Yes, it's on. Expect the crowd from about 6pm."
- **Every line is labelled and cited**
  Fact Inference Assumption Unknown with the Ticketmaster listing, MSG's page or Wikidata as sources.
- **It adapts**
  When Ticketmaster's page refused access, it switched to MSG's official page instead of guessing.
- **The steps are on record**
  Every tool call is logged, so anyone can audit how it got there.

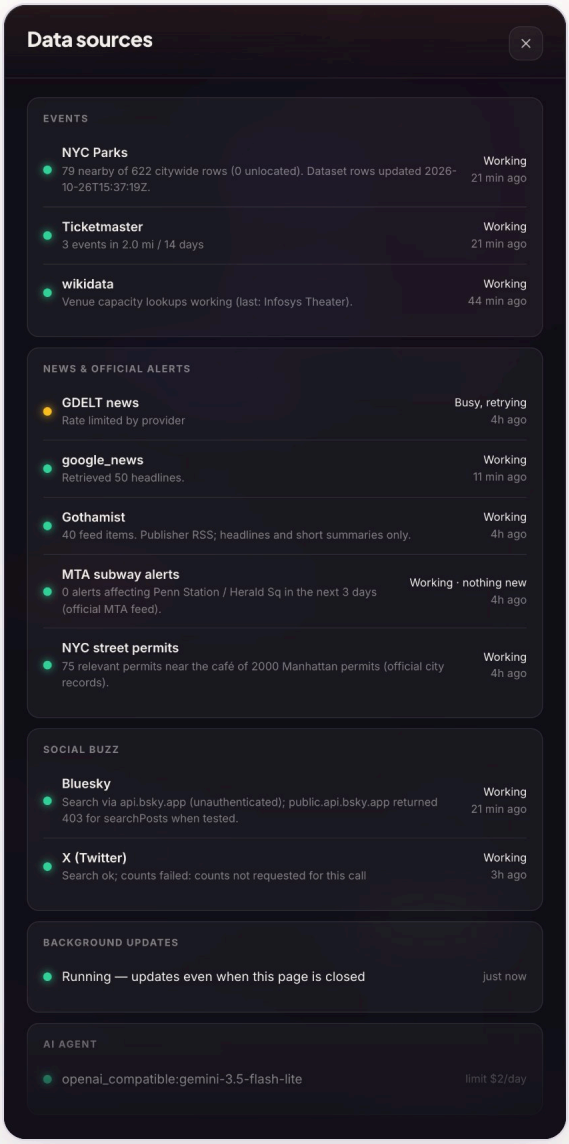


AGENT **Follow a question.** Maya writes it once ("Let me know if the Rangers game is called off"). The agent checks right away with the listing, the news and the event's history, gives a verdict with reasons, re-checks every 6 hours on its own, and alerts her only when the answer changes. Repeats and opinion pieces don't set it off.

6 TRUST AND SAFETY

Why you can believe it

An AI that gives business advice has to be honest about what it knows. These rules are built into the product, not left to chance.



TRUST **Data sources.** Live health of every source, when it last worked, and the AI model in use with its daily cost limit.

Every number has a source

Crowd figures, distances and money come from code and cited data. If a number is an estimate, it says “EST” and explains the method.

Disagreement is shown

If two sources disagree, the claim is marked “disputed”, not quietly picked. An unconfirmed rumour leads to an explicit “no change”.

Only approved sources

Official APIs, an allow-list of websites and robots.txt respected. If a page blocks us, we report it; we don’t get around it.

A human decides

Nothing is posted, sent or bought by the app. Plans wait for Maya’s review. Paid data runs only under a cap she set.

7 HOW IT MEETS THE ASSIGNMENT

The Newsren plan, mapped

The Newsren plan proposes six features and one rule that isn't optional: real agentic capability. LocalLift takes that blueprint and applies it to one specific reader.

NEWSREN FEATURE	IN LOCALLIFT	STATUS
My Newsren personal research agent	The Local Briefing for this café's location, hours, menu and budget, plus the research agent (Investigate and Ask AI)	Built
Story Intelligence	Developments group news, official records and listing changes into one evolving story with a "what changed?" history	Built
News-to-Learning audience adaptation	The same evidence rewritten for the owner (briefing), the staff (Rush Radar briefing) and customers (sign and social copy)	Built
Industry Impact	"Why it matters to Opening Act": impact on hours, capacity and demand, plus profit scenarios worked out by code	Built
Evidence & Bias Inspection	Source types, claim labels, checks against the official listing, disputed claims kept, a news relevance rule	Built
Follow This Story	Follow a question: the agent re-checks every 6 hours and alerts only when the verdict flips	Built
Agentic core	A goal per run, 14 typed tools, budgets and validation, plans and adapts, reports what it couldn't verify	Built
Briefing delivery	Opt-in email delivery of briefings, with approval first	Next step

What we'd build next

- Email or text delivery of the morning briefing, with the owner's approval.
- Learning from real outcomes: compare predicted walk-ins with recorded sales to tune the estimates.
- More neighbourhoods and business types: the same engine works for a bar near a stadium or a shop near a theatre.

> **The news already knew. Now the café does too.**

LocalLift finds what's coming, checks it, explains why it matters and suggests one clear next step, while a person keeps the final say.

8 HOW IT'S BUILT

The tech stack, and what each part does

Simple, well-known tools, chosen so the whole thing runs on a laptop and costs nothing beyond an optional $5 a month for X.

LAYER	TECHNOLOGY	WHAT IT DOES HERE
Backend & API	Python 3 · FastAPI · Uvicorn	Serves the web app and every API call: briefing, events, plans, projections, questions
Database	SQLite	Stores events, news, developments, claims, plans, outcomes and the agent's run logs
Background worker	Python scheduler	Refreshes sources, re-checks followed questions every 6 hours, runs the daily X sample
AI agent	Gemini 3.5 Flash-Lite · Groq Llama 3.3 (backup)	Our own tool-use loop with 14 typed tools, budgets and validation. Also supports Claude.
Events	Ticketmaster Discovery API · NYC Parks Open Data	Concerts, games, shows and park events within 2 miles
News & official alerts	Gothamist RSS · Google News RSS · GDELT · NYC street permits · MTA alerts	Local stories, event coverage, street closures and subway changes
Social buzz	Bluesky API · X API v2	Post samples and volume trends; X capped at $5 a month
Crowd estimates	Wikidata	Venue capacities, checked for errors
Frontend	HTML, CSS, JavaScript · Leaflet + OpenStreetMap · Lucide icons	The dark, animated interface, the neighbourhood map and icons; no framework needed
Testing	pytest · Playwright	78 automated tests on fixed test data; Playwright took this document's screenshots
Code & hosting	GitHub · GitHub Actions · Vercel	Source control, the daily data refresh, and the live prototype (auto-deploys on every push)

TRY THE LIVE PROTOTYPE
locallift-pied-mu.vercel.app

READ THE CODE
github.com/sumeethaldipur/LocalLift

The online prototype's data is refreshed every morning by a GitHub Actions job that runs the same background jobs as the laptop version.